

HA NGUYEN, Ph.D.

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SUMMARY

Principal Network Systems R&D Architect and Hardware Engineer (Ph.D.) with 8+ years of combined post-doctoral R&D and research leadership bridging scale-out network transport protocol design with physical FPGA silicon acceleration. Recognized for a versatile technical profile capable of operating across traditional functional boundaries: leading dedicated software/protocol architecture groups, embedded RTL design teams, or cross-functional HW/SW Co-Design divisions. On the networking side, specializes in low-latency transport protocols (TCP/NC, RoCE/RDMA), P4 programmable data planes, active queue management (AQM), dynamic SDN routing, and discrete-event C++/Python network modeling in ns-3. On the hardware side, offers expertise in SystemVerilog RTL, ultra-wide 1024-bit AXI4-Stream datapaths, PCIe 3.0 DMA offload engines, high-speed DSP pipelines, and multi-vendor FPGA emulation across Intel Agilex/Stratix and AMD Xilinx Kintex UltraScale+ platforms. Backed by 40+ peer-reviewed publications (IEEE/IEICE) and \$100k+ in competitive grants, excels at transforming complex protocol mathematics into high-performance production hardware and software architectures.

TECHNICAL SKILLS

Core Domains : HW/SW Co-Design, Scale-Out Network Architectures, Protocol Stack Design, Hardware Acceleration, Discrete-Event Network Modeling, Pre/Post-Silicon Validation, High-Speed Datapaths, Hardware-in-the-Loop (HITL) Emulation, Active Queue Management (AQM), In-Band Network Telemetry (INT), Packet Loss Masking, Traffic Engineering

Silicon & Hardware : SystemVerilog, Verilog, Intel Agilex 5/7 & Stratix II/III/IV, AMD Xilinx Kintex UltraScale+, AXI4-Stream, APB Bus, PCIe 3.0 DMA, SerDes (156.25 MHz), Control/Status Registers (CSRs), Fixed-Point Arithmetic (10.22 Format), Altera DSP Builder, Synopsys Symphony, Vivado, Quartus, ModelSim

Networking & Protocols : Transport Protocols (TCP/IP, TCP/NC, UDP, RoCE/RDMA), Programmable Data Planes (P4-16, BMv2, P4Runtime/gRPC), Software-Defined Networking (SDN / OpenFlow, Ryu Controller), Network Coding (NC), Power Line Communication (PLC IEEE 1901.1), Active Queue Management (AQM), Erasure Coding (EC), TDMA Scheduling

Software & Tools : C++, Python, ns-3 Simulator, Mininet, TCL Automation, Linux (Ubuntu, Embedded), MATLAB/Simulink, Git, Wireshark, Scapy, Linux tc

EXPERIENCE

University of Science

Oct. 2021 – Apr. 2026

Network Systems R&D Researcher & Laboratory Supervisor

Ho Chi Minh City, Vietnam

- Directed major R&D initiatives within the ICTLAB bridging scale-out transport protocol design, programmable P4 data-plane telemetry, active queue management (AQM), and high-throughput hardware/software co-design.
- Architected legacy-transparent middlebox transport proxies and proactive loss-masking queue models, enabling multi-session flow aggregation over heterogeneous networks without requiring host OS kernel modifications.
- Co-founded the laboratory infrastructure; managed full-lifecycle hardware prototyping, C++/ns-3 discrete-event network modeling, and APB/AXI memory-mapped register interface development.
- Secured and executed over \$100,000 in competitive industry and government R&D grants across 7 projects (including VNU-HCM Class-B Grant B2023-18-01), publishing 20+ peer-reviewed papers in top IEEE/IEICE venues.

Kyushu Institute of Technology

Apr. 2018 – Sep. 2021

Principal Postdoctoral Researcher & Principal Investigator (PI)

Fukuoka, Japan

- Served as Principal Investigator (PI) for two prestigious JSPS KAKEN Grants-in-Aid (Grant IDs: 18H06467 & 20K19785, totaling 4.42M JPY), leading 3+ years of post-doctoral R&D on resilient network protocol architectures.
- Engineered fault-tolerant transport state machines and dynamic SDN control logic, deploying active loss masking and adaptive ACK mechanisms to eliminate TCP timeout stalls and suppress tail latency.
- Developed modular discrete-event simulation models in C++ and Python to validate real-time queue depth tracking, dynamic buffer management, and bandwidth fairness under heavy burst loss.
- Designed and delivered technical guest lectures on SDN control mechanics, OpenFlow abstractions, and Mininet programmable network topologies for the City College of New York (CCNY).

University of Science

Sep. 2009 – Mar. 2014

Associate Lecturer & Departmental Network Administrator

Ho Chi Minh City, Vietnam

- Conducted applied physical-layer hardware R&D, architecting a 2×2 MIMO-OFDM SDM baseband transceiver on FPGAs achieving a low-latency 400 Mbps forward pipeline using custom DSP units.

- Administered departmental enterprise networking infrastructure, managing L2/L3 switching, OSPF/BGP routing protocols, VLAN segmentation, and firewall security in live production environments.
- Maintained active certifications as a CCAI (Cisco Certified Academy Instructor), mentoring student teams to a 2nd-place national finish in the Cisco NetRiders Competition (2013).

MASTER PROJECT REPOSITORY

High-Speed Hardware-in-the-Loop (HITL) Link Emulation

Apr. 2026 – Present

- **Tools:** Intel Agilex 5 (DE25), Xilinx Kintex UltraScale+ (KU5P), TI LMK62E2, SystemVerilog, PCIe 3.0, DMA, HSMC, SFP-to-SMA.
- Architected a heterogeneous hardware-in-the-loop (HITL) system bridging Intel Agilex 5 and AMD Xilinx Kintex UltraScale+ FPGAs across high-speed multi-vendor hardware interfaces.
- Integrated a TI LMK62E2 high-performance oscillator delivering a 156.25 MHz SerDes reference clock to lock transceivers and synchronize high-speed physical link layers.
- Designed an ultra-wide 1024-bit SystemVerilog datapath operating on a 156.25 MHz / 312.5 MHz core clock (achieving up to 320 Gbps internal pipeline capacity) for real-time header parsing, deparsing, and packet filtering.
- Engineered host-to-FPGA streaming infrastructure via PCIe 3.0x4 DMA engines, enabling host interaction with embedded Network Coding (NC) hardware offload logic.

Wireless System Emulation: Multi-User AWGN Hardware Generator

Sep. 2025 – Feb. 2026

- **Tools:** SystemVerilog, Intel Agilex 7, Xilinx UltraScale+, Vivado, Quartus, ModelSim, APB Bus, TCL, C/C++ Drivers.
- Architected a resource-optimized, real-time embedded Gaussian noise (AWGN) generator block in SystemVerilog, integrated with an APB bus interface for memory-mapped register access (CSRs) by C/C++ host software.
- Designed 512-bit wide AXI4-Stream datapaths achieving 375 MHz+ operating frequencies on Intel Agilex 7 and Xilinx FPGAs to meet strict pre-silicon timing requirements.
- Automated FPGA project creation, IP synthesis, and pin mapping by authoring modular TCL build scripts, streamlining continuous integration and verification flows.

Programmable Data Plane (P4) for In-Band Telemetry, AQM & Loss Recovery

Jan. 2024 – Jan. 2025

- **Tools:** P4-16, BMv2 Switch, P4Runtime / gRPC, Mininet, ns-3, Scapy, Linux tc.
- Programmed custom P4 datapaths featuring In-Band Network Telemetry (INT) with 8-byte custom headers to track packet arrival jitter, link loss rates, and queue depth in real time at line rate.
- Engineered a QoS-aware selective recirculation port queue strategy in P4 to buffer packets at intermediate switches, enabling localized retransmissions that reduced 1 GB multicast transfer times by 36% across 41 nodes.
- Implemented an adaptive queue depth mechanism that dynamically shifts congestion thresholds based on real-time occupancy, cutting packet loss by 59% under 300 Mbps traffic loads and accelerating recovery times by 67%.

Fault-Tolerant TDMA Framework for LWSNs & Constellation Networks

Oct. 2023 – Sep. 2025

- **Tools:** Network Simulator 3 (ns-3), C++, Python, Erasure Coding (EC), TDMA Scheduling.
- Architected a proactive loss recovery framework combining hop-by-hop Erasure Coding with TDMA scheduling for multi-hop wireless networks, evaluating against reference hardware profiles (ARM Cortex-M3 / Atmel AT86RF215).
- Engineered a 1-step Hill Climbing search heuristic that computes near-optimal timeslot allocations in 1.32 ms for 18-node topologies under strict battery energy budgets.
- Achieved >0.99 delivery probability across a 30-node topology under severe 0.5 link loss rates, maintaining node energy profiles capable of 10-day operation on a 1000 mAh battery.

Self-Healing Dynamic Routing Protocol for SDN (SDN-FEDRP)

Jan. 2021 – Dec. 2021

- **Tools:** OpenFlow, RYU Controller, Mininet, Python.
- Architected an adaptive dynamic routing protocol for Software-Defined Networks (SDN) to eliminate traffic hot-spots, balance real-time link utilization, and prevent flow congestion.
- Engineered a custom routing algorithm on the Python-based RYU controller leveraging Depth First Search (DFS) for dynamic path discovery and link cost recalculation.
- Implemented an automated OpenFlow flow re-allocation mechanism governed by a fairness factor, achieving a 7x load balancing performance improvement over standard SDN-OSPF in Mininet tests.

Resilient Transport Protocol Suite & State Machine Architecture

Mar. 2015 – Mar. 2020

- **Tools:** Network Simulator 3 (ns-3), C++, Python, Power Line Communication (PLC IEEE 1901.1), Gilbert Loss Model.
- Engineered advanced transport layer state machines incorporating Forward Retransmission (FR), Dependence Retransmission (DR), and a custom 32-bit Packet Loss Sequence (PLS) header field for accurate loss tracking.
- Designed a Dynamic ACK-Skipping scheme with ACK Accumulation and a 200 ms Timer-Based ACK Retransmission engine to maintain high goodput under severe bi-directional loss rates up to 0.4.
- Implemented a source-side Duplicate ACK Generator that locally synthesizes proxy DupACKs to trigger fast TCP

congestion recovery without waiting for full end-to-end round-trip times.

- Devised a Pause-ACK Buffer leveraging Packet-ID (Pid) delta tracking to delay out-of-order ACKs, eliminating unwarranted TCP congestion window reductions under path delay variations up to 2.0x.
- Developed an Adaptive Mid-Window Encoding Matrix that recalculates Network Coding coefficients on-the-fly based on Gilbert loss model estimation (reducing RMSE to 0.74) without forfeiting active coding window data.

Transparent Middlebox Overlay Tunneling & AQM Buffer Management

Jun. 2017 – Dec. 2019

- **Tools:** ns-3, C++, Python, Linux Networking.
- Formulated a middlebox encapsulation proxy architecture aggregating multiple independent TCP flows across asymmetric links (1–7 Mbps) into a single optimized stream with only 4% encapsulation overhead.
- Achieved zero-touch legacy transparency, requiring zero OS kernel or driver alterations to end-user devices.
- Designed a state-based buffer management algorithm applying 4 distinct timeout triggers to purge "dead packets," saving memory footprint and preventing mistaken redundancy escalation.

FPGA-Based MIMO Channel Emulators, Transceivers & Turbo Codes

Sep. 2011 – Nov. 2015

- **Tools:** SystemVerilog, Verilog, Quartus, Synopsys Synphony, Altera DSP Builder, Stratix II/III/IV, MATLAB/Simulink.
- Architected a 4×4 MU-MIMO real-time channel emulator in SystemVerilog; optimized matrix DSP pipelines to reduce Stratix FPGA ALM resource consumption from 16,800% down to 20% while downsampling to 2604 Hz.
- Implemented a 2×2 MIMO-OFDM Spatial Division Multiplexing (SDM) baseband transceiver achieving a 147.12 MHz evaluation clock speed on Stratix III FPGAs while consuming $< 45\%$ dedicated logic registers.
- Designed a Parallel Concatenated Convolutional Code (PCCC) Turbo Encoder and Soft Output Viterbi Algorithm (SOVA) decoder on FPGA, optimizing metric updates by tracking a single ML survivor route to reduce computational complexity.
- Maintained model-to-silicon arithmetic accuracy by utilizing a 32-bit fixed-point (10.22 format) representation, resulting in a hardware Bit Error Rate (BER) performance that exactly matched software models.

PUBLICATIONS, GRANTS & AWARDS

- **Peer-Reviewed Publications:** Authored 40+ international papers (IEEE/IEICE/Scopus), including 7 journal articles, focusing on transport layer optimization, P4 programmable data planes, loss-masking proxies, and FPGA DSP pipelines.
- **Competitive R&D Grants:** Principal Investigator (PI) and Key Lead across 10+ competitive research grants in Japan (JSPS KAKEN 18H06467 & 20K19785) and Vietnam (VNU-HCM B2023-18-01), securing over \$100,000+ in total funding.
- **Awards & Honors:** Recipient of 3 International Conference Best Paper Awards (2019, 2020, 2022), Cisco Instructor Service Award (Mentored 2nd-Place Cisco NetRiders National Student Team, 2013), and multiple Academic Excellence Awards.

CERTIFICATIONS

- **Cisco Systems:** CCNA, CCNA Security, CCAI (Cisco Certified Academy Instructor)
- **Microsoft:** MCSE (Systems Engineer) & MCSA (Systems Administrator) – Network Infrastructure

EDUCATION

Kyushu Institute of Technology

Ph.D. in Computer Science and Systems Engineering

Apr. 2014 – Mar. 2017

Fukuoka, Japan

University of Science

M.S. & B.S. in Electronics and Telecommunications Engineering

Sep. 2005 – Sep. 2012

Ho Chi Minh City, Vietnam